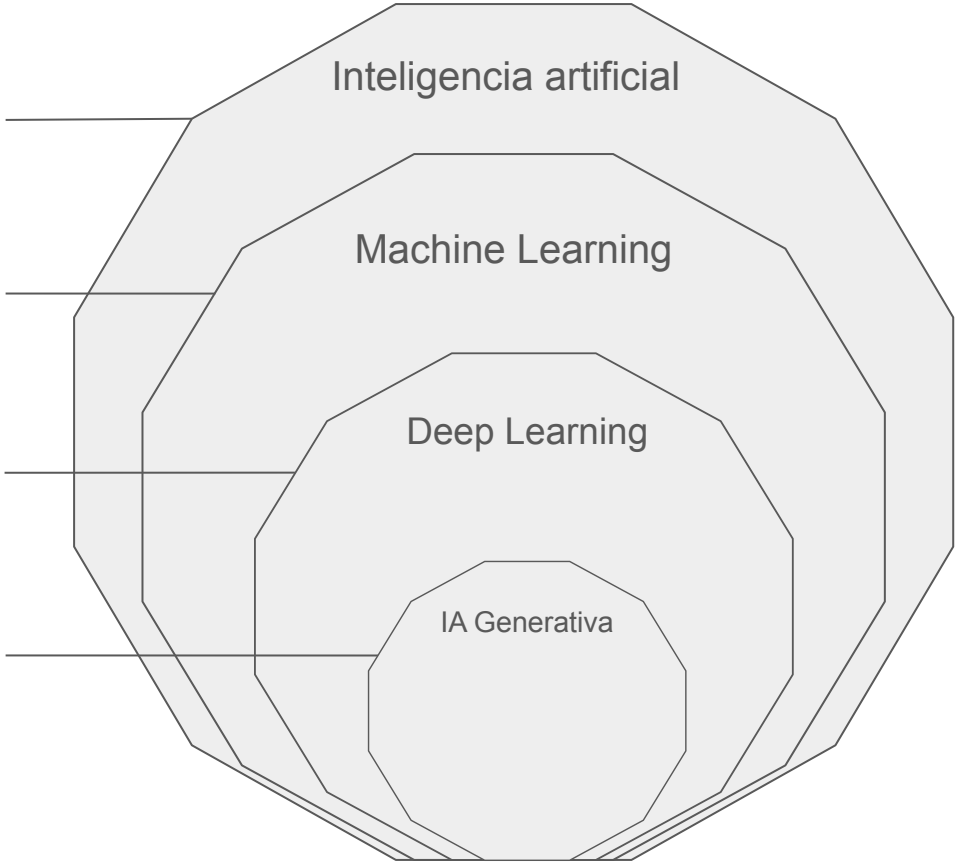
¿Qué es ChatGPT?

1956 El esfuerzo para automatizar tareas intelectuales normalmente realizadas por humanos.

90’s Subconjunto de la IA que permite a las máquinas aprender patrones de datos existentes para tomar decisiones o hacer predicciones sin ser programado explicitamente para cada caso

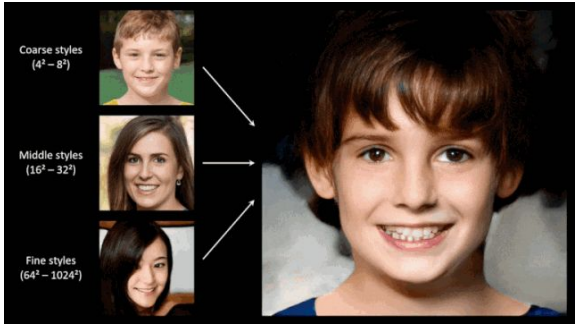
2012: Una técnica de aprendizaje automático en la que se aprenden interacciones más complejas incluso sin que nosotros le digamos qué buscar

2021: Crea nuevo contenido escrito, visual y auditivo a partir de indicaciones o datos existentes.



Dentro de la IA Generativa…

GANs

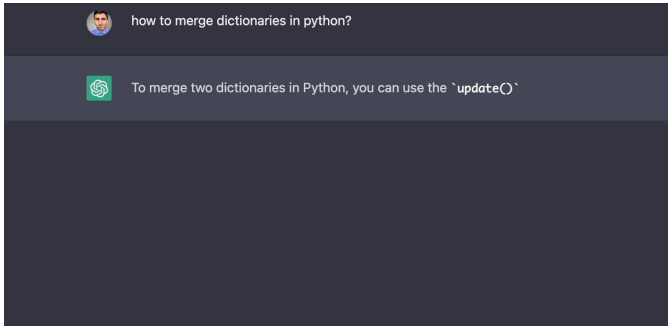


StyleGAN
Pix2pix

Diffusion model

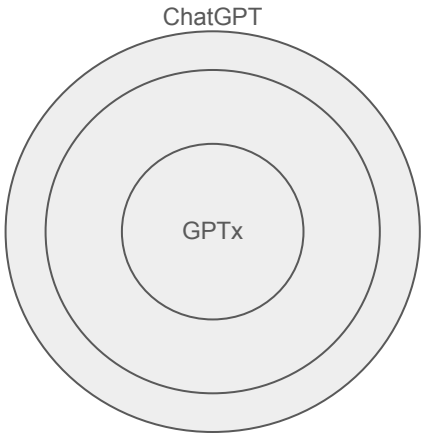
Dall-e
Midjourney

LLM

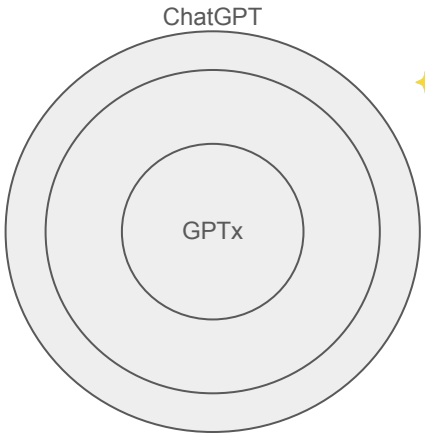


GPT, DeepSeek,
Llama, Gemini, Mistral

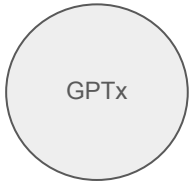
¿Qué es ChatGPT?



¿Qué es ChatGPT?

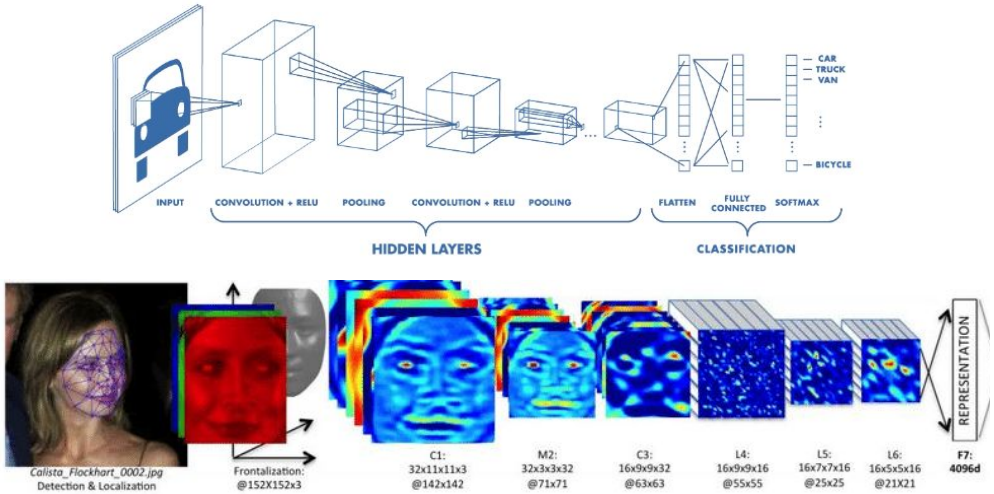


✨Large Language Models✨



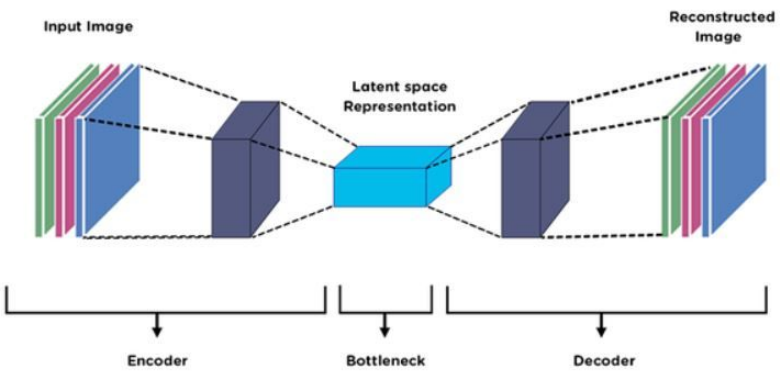
De la CNN a una LLM

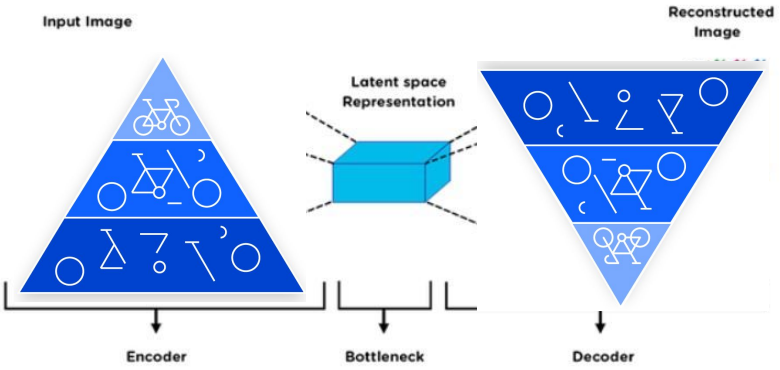
Arquitectura de una red convolucional:



De la CNN a una LLM

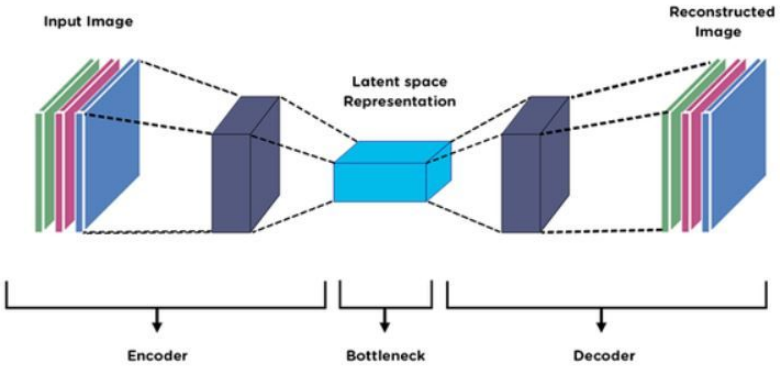
Arquitectura Auto-encoder:





De la CNN a una LLM

Arquitectura Auto-encoder:



Arquitectura de un transformer:

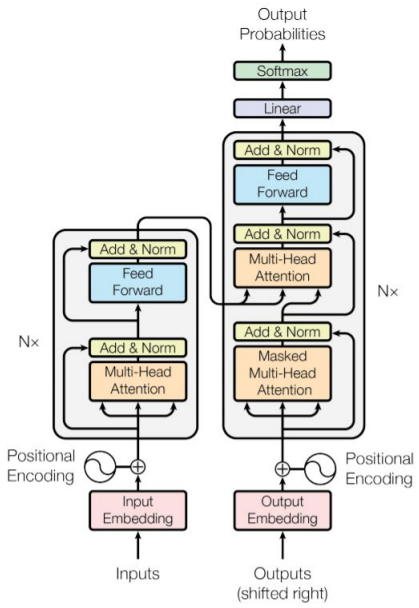
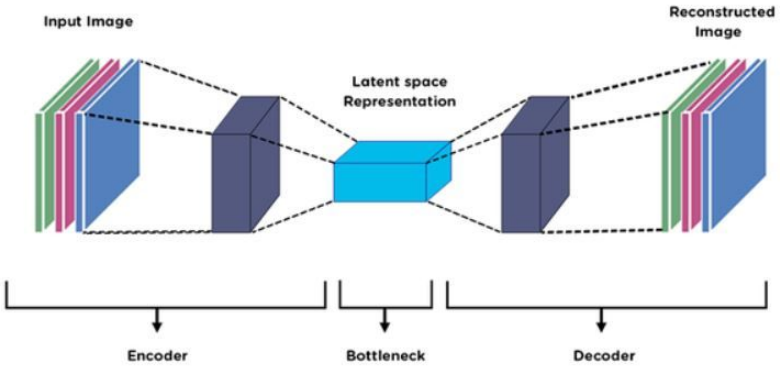


Figure 1: The Transformer - model architecture.

De la red neuronal simple a una LLM

Arquitectura Auto-encoder:



Arquitectura de un transformer:

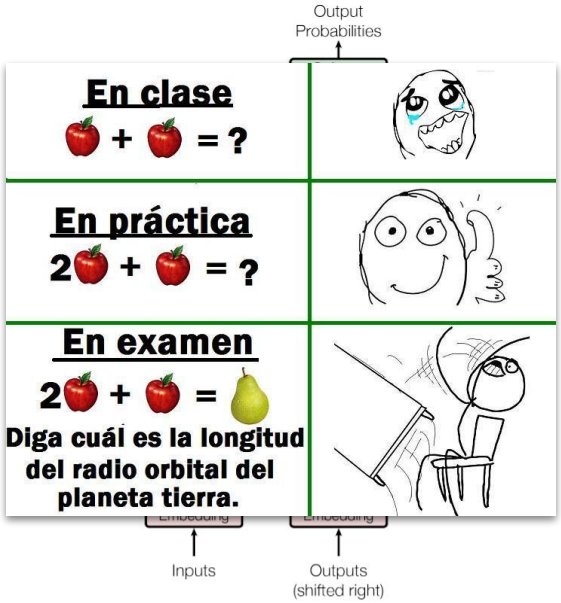
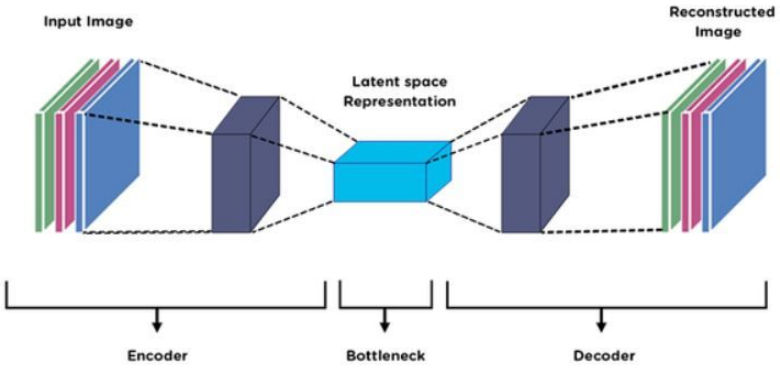


Figure 1: The Transformer - model architecture.

De la red neuronal simple a una LLM

Arquitectura Auto-encoder:



Arquitectura de un transformer:

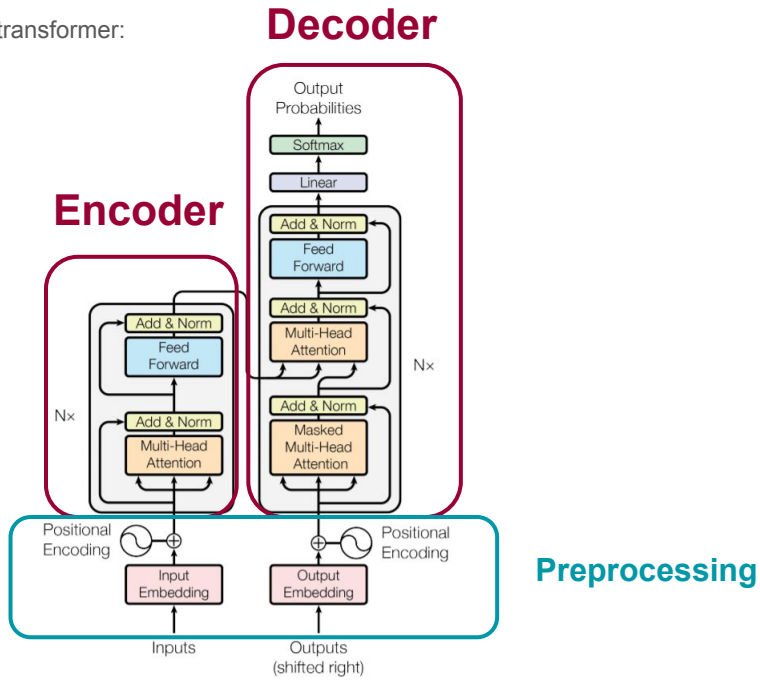
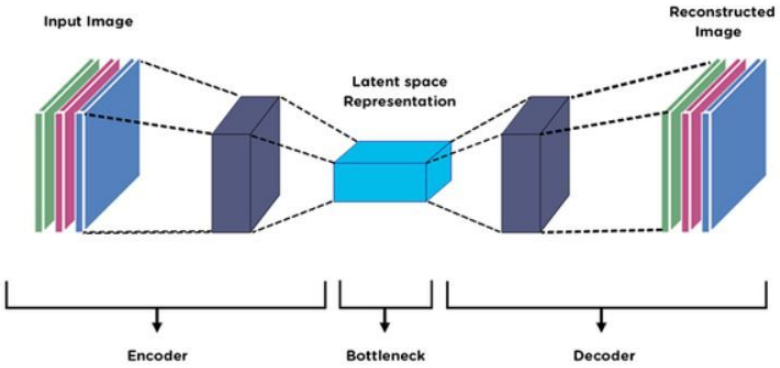


Figure 1: The Transformer - model architecture.

De la red neuronal simple a una LLM

Arquitectura Auto-encoder:



Arquitectura de un transformer:

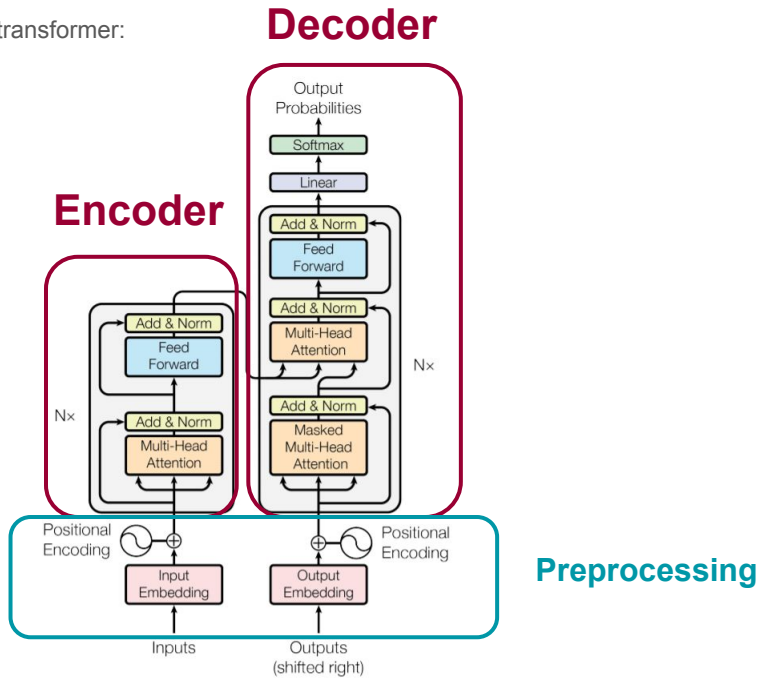
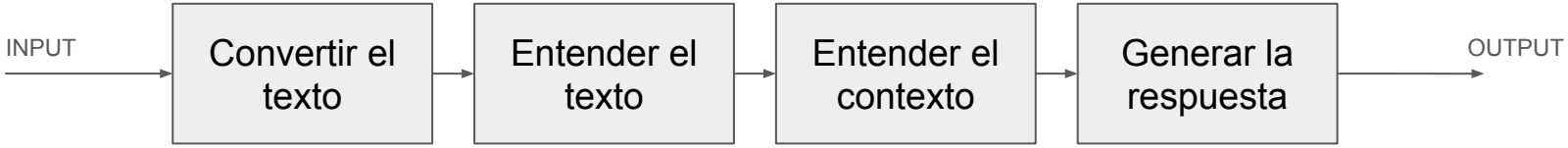
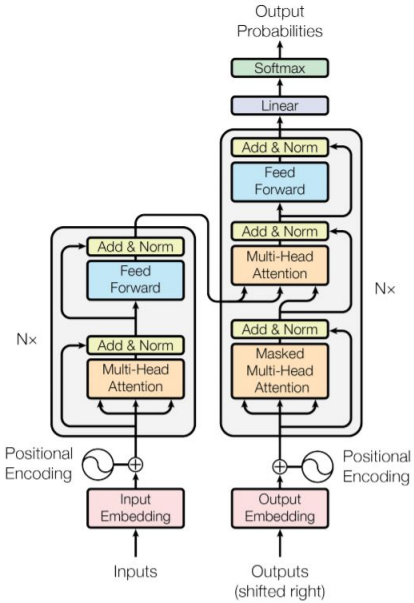


Figure 1: The Transformer - model architecture.

Pero…¿Cómo funciona?





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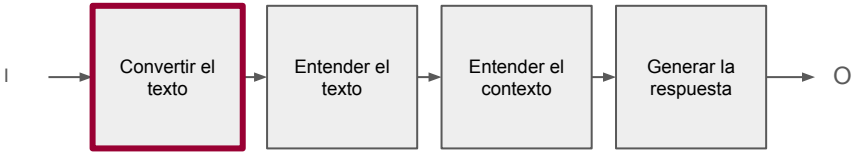
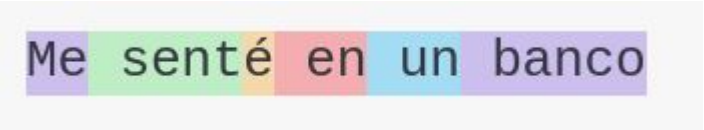


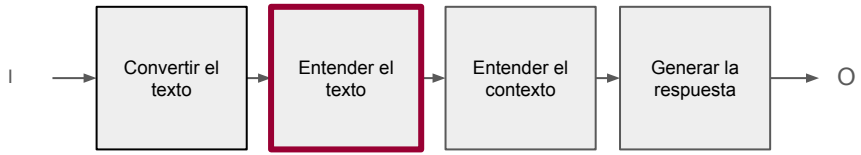
Tokenización

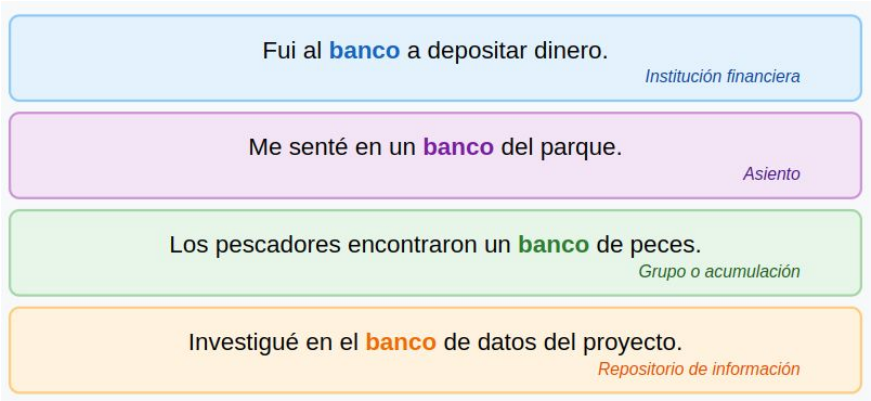
Proceso por el cual las palabras se convierten a un lenguaje que la máquina pueda entender. Llamamos token a la unidad básica en la que está representada una palabra

Me senté en un banco del parque



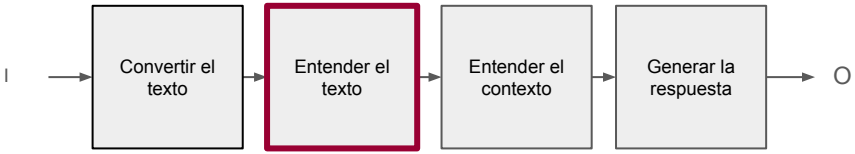
Pero…¿Cómo funciona?





Dependiendo del contexto, la palabra banco toma diferentes significados, necesitamos una representación numérica que pueda describir este fenómeno.

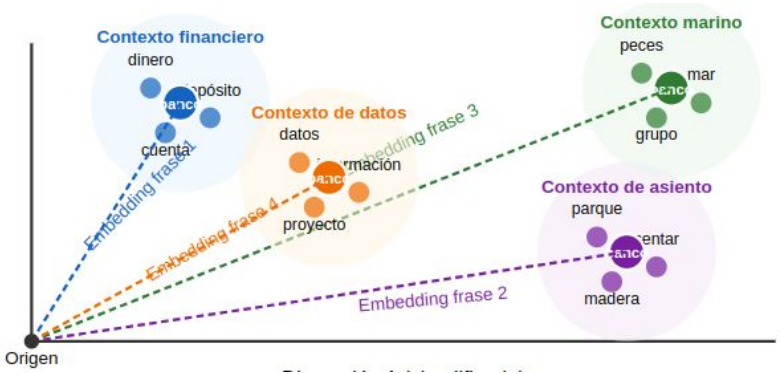
Pero…¿Cómo funciona?



Embedding

Un embedding es un vector que describe la relación entre tokens creando la representación de la totalidad de un vocabulario

1. Me → [1,0,0,0,0]
2. senté → [0,1,0,0,0]
3. en → [0,0,1,0,0]
4. un → [0,0,0,1,0]
5. banco → [0,0,0,0,1]



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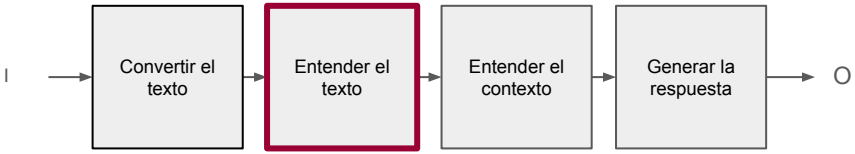

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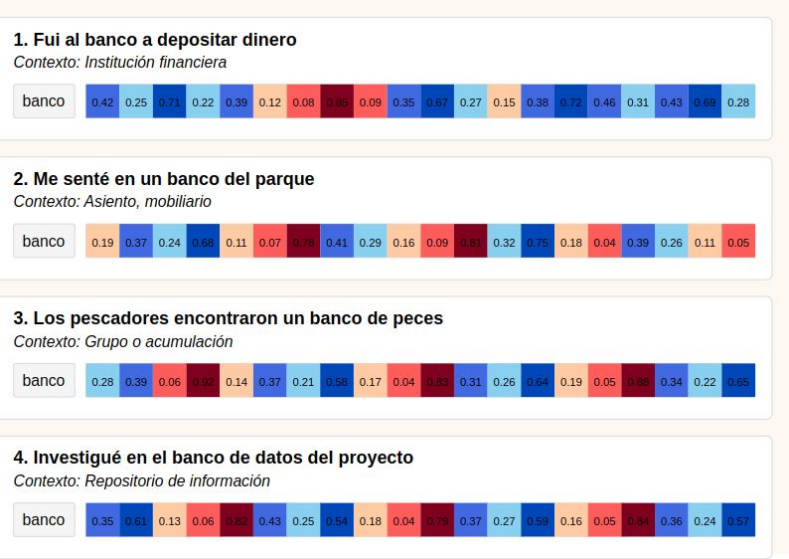

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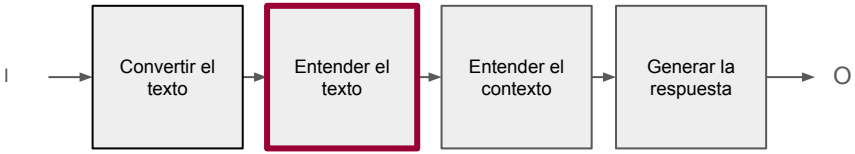

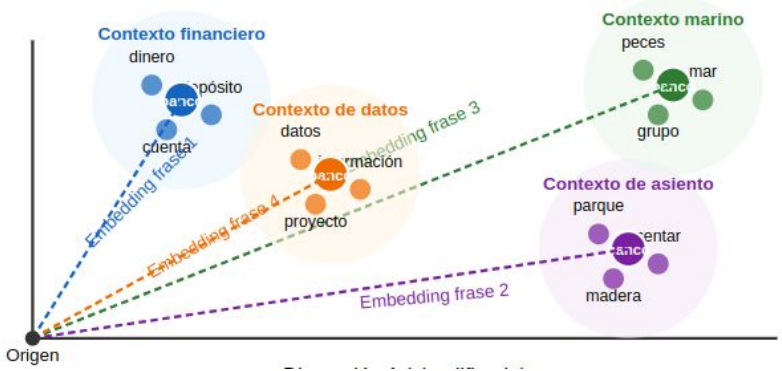

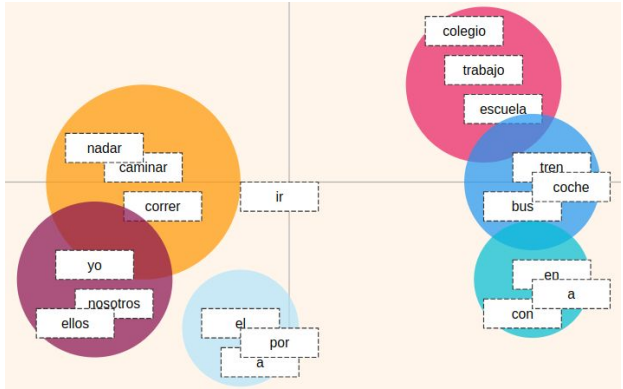

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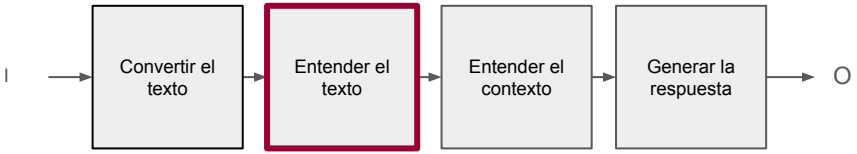
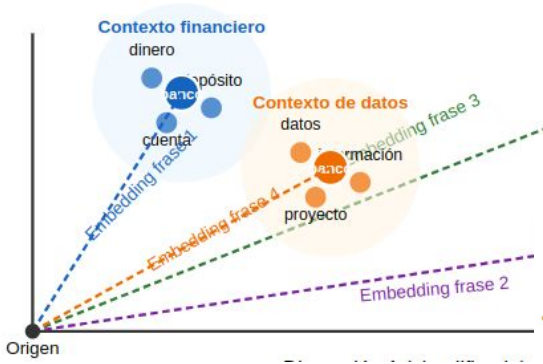
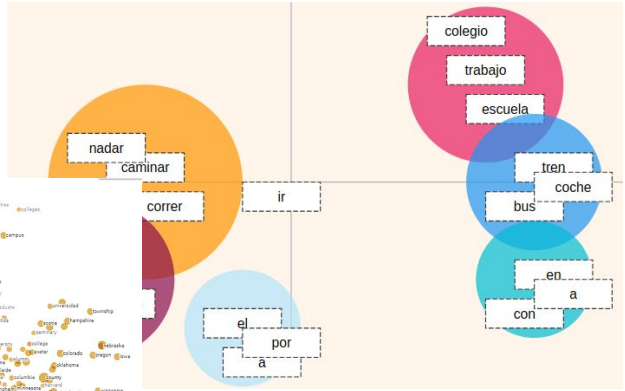
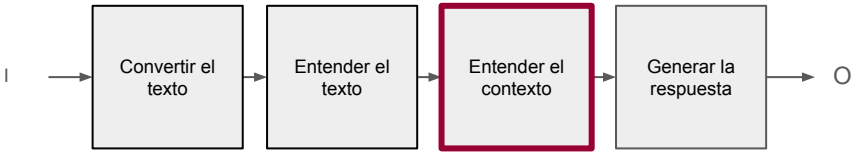


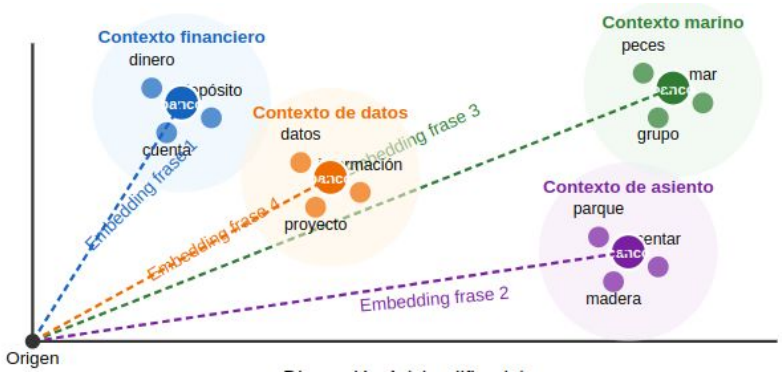


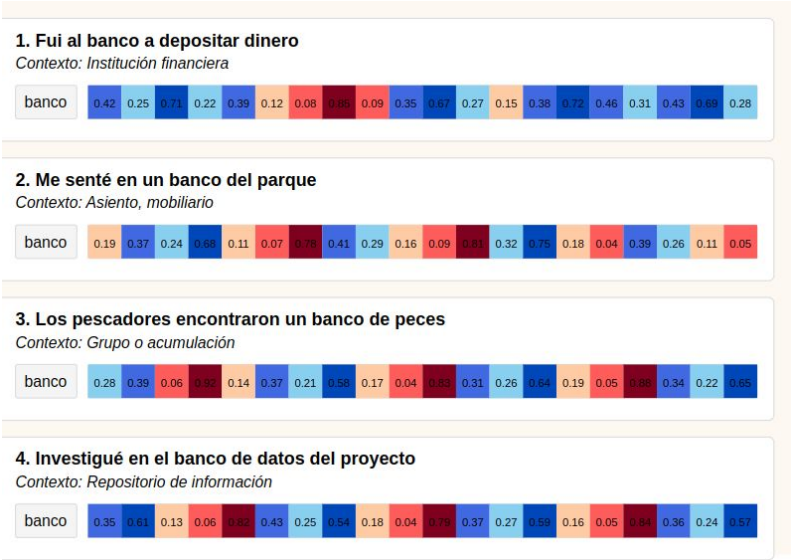


Pero…¿Cómo funciona?

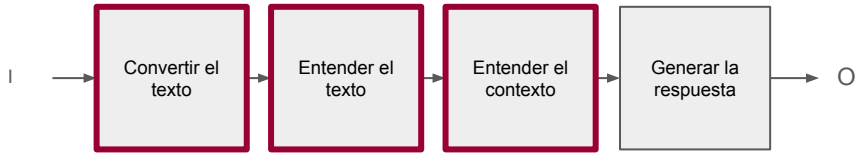


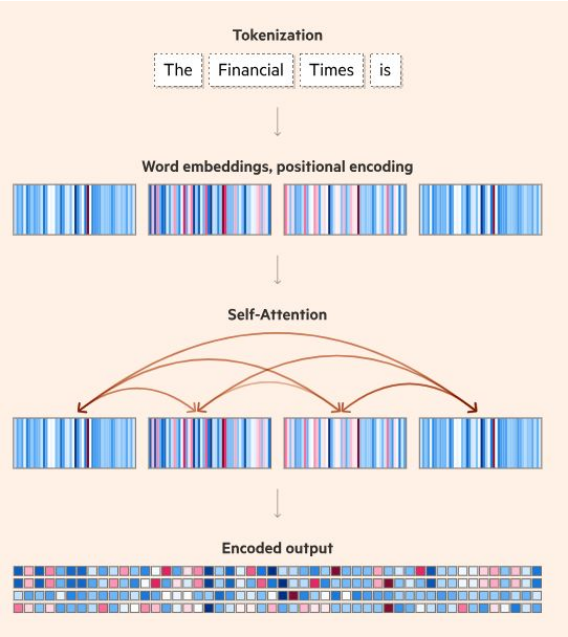
Bloque de atención: self attention





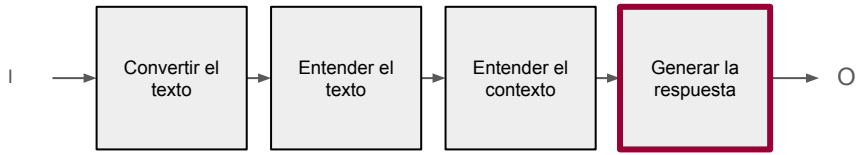
Pero…¿Cómo funciona?

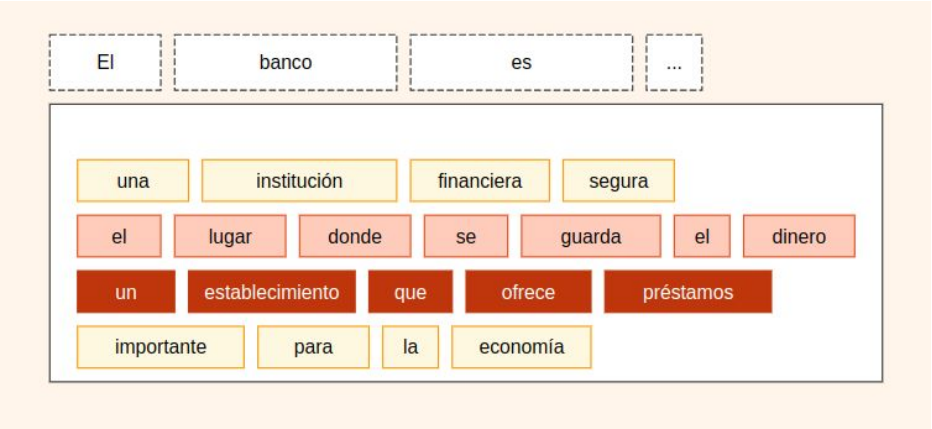




En este punto lo que tenemos es una representación de nuestro input que la máquina entiende, incluido el significado, la posición y la relación entre palabras.

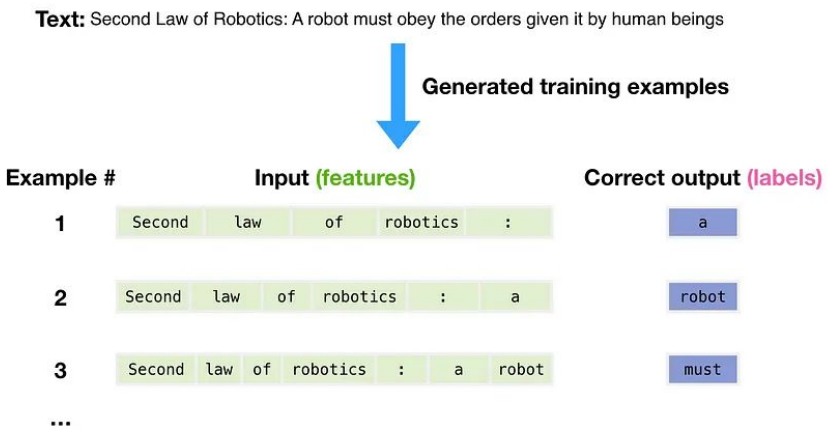
Pero…¿Cómo funciona?

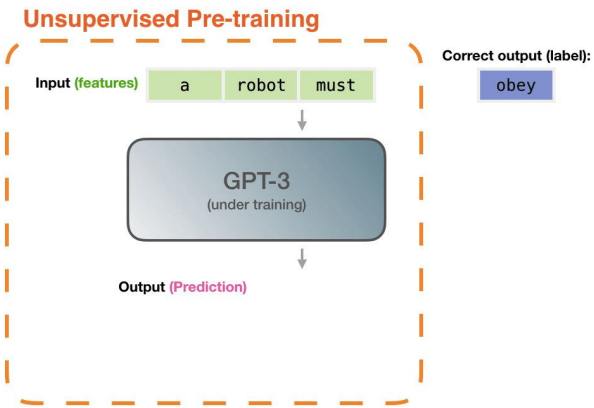




Para generaeliger la respuesta, el modelo da una puntuación probabilística para un conjunto de tokens. Así, explora múltiples vías hasta que la mejor “ruta” que representa la cercanía de ser el siguiente conjunto de palabras en la secuencia, esto se repite continuamente hasta que el modelo está satisfecho con el texto producido.

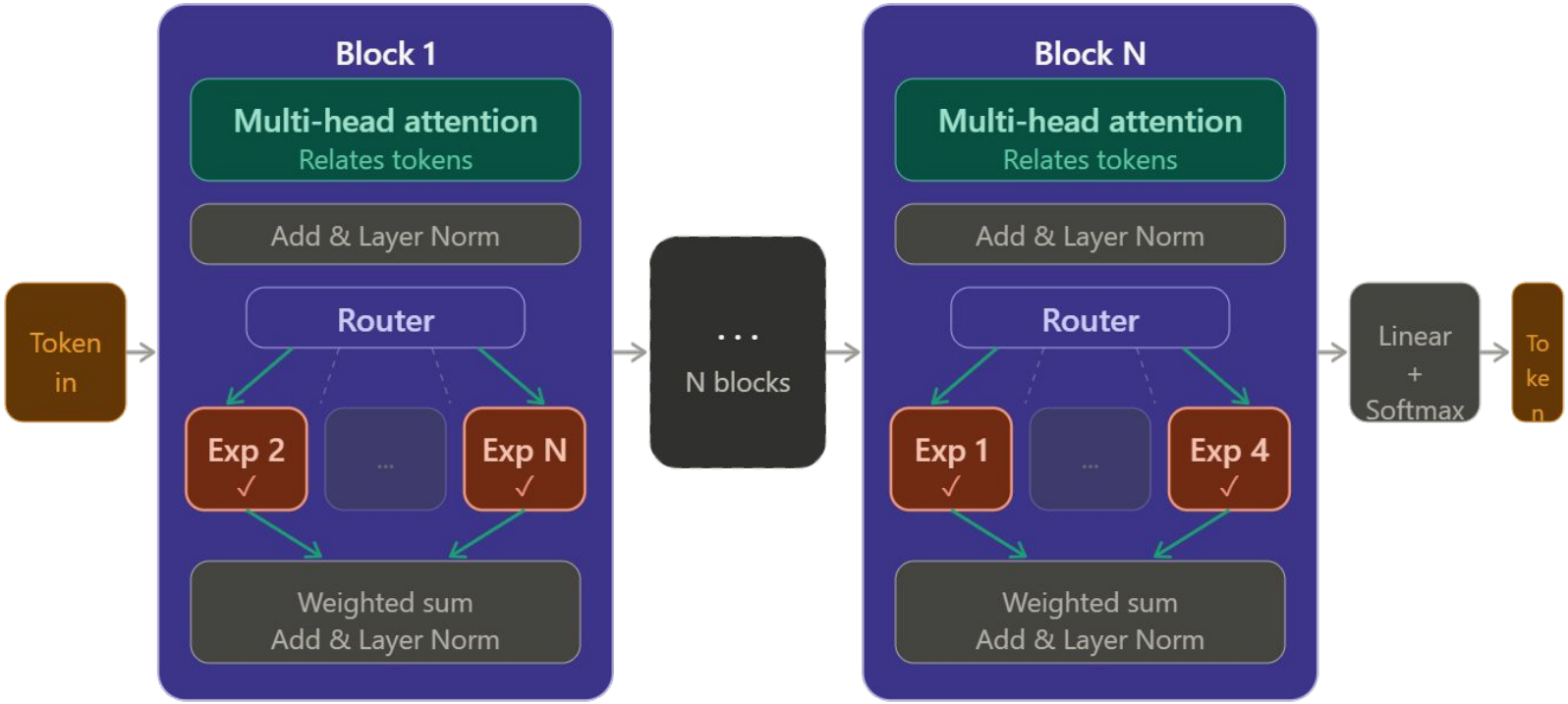
Y… ¿cómo se entrena?



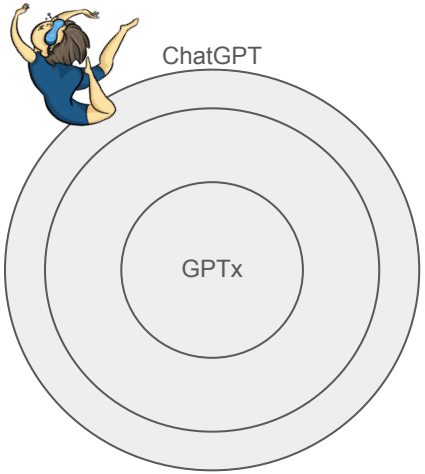


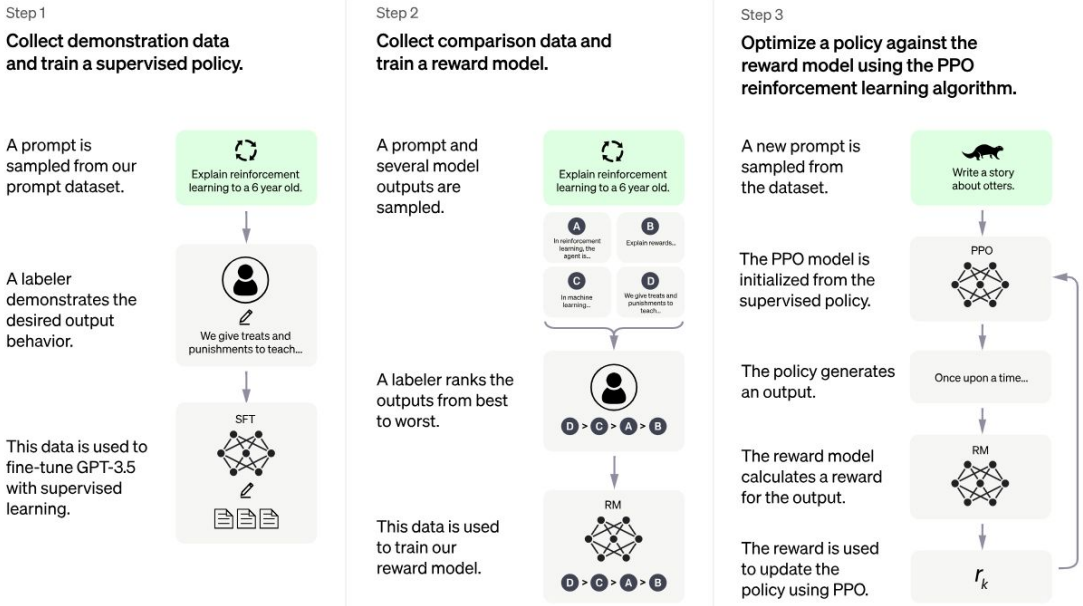
DUDAS

Mixture of Experts (MoE)



Volviendo a chatGPT





Desmitificando a las LLMs

🤖

¿Qué son realmente?

Predictores de texto muy sofisticados que "adivinan" la siguiente palabra más probable basándose en patrones.

⚠️

Limitaciones importantes

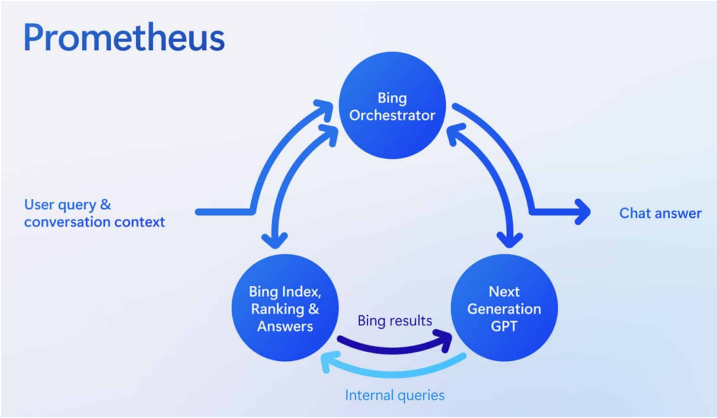
- Pueden "alucinar" información falsa
- Conocimiento con fecha de corte
- No "entienden" como humanos

🎯 **Regla de oro:** Úsalos para tareas donde "aproximadamente correcto" es suficiente, no para información exacta

Claude vs Bing AI vs Perplexity AI

Bing AI

perplexity



Can u please talk me about the hardware and energy cost of training GPT-4

Sources

The carbon footprint of GPT-4 towardsdatascie... • 1

What is GPT-4? Everything You Need to... techtarget • 2

GPT-4 details leaked reddit • 3

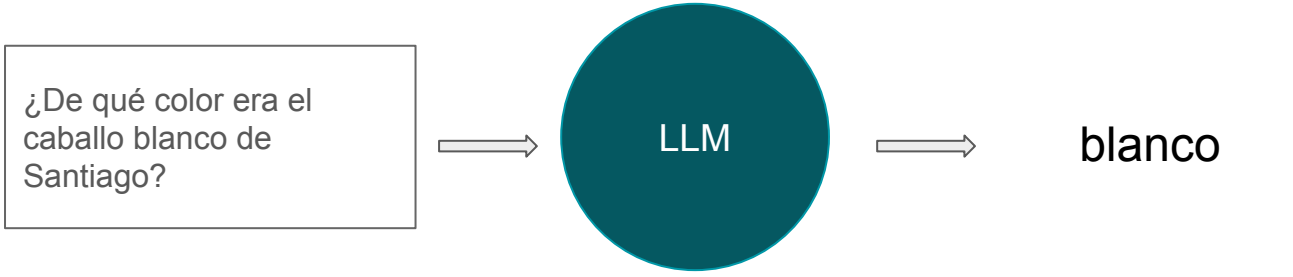
View 2 more

Answer

The estimated energy cost of training GPT-4 is significant. According to leaked information, GPT-4 was trained on about 25,000 Nvidia A100 GPUs for 90-100 days, which translates to a substantial amount of electricity consumption [1]. The exact energy consumption is not disclosed, but the training process of its predecessor, GPT-3, required 3.2 million USD in computing resources alone, which involved running the model on 285,000 processor cores and 10,000 graphics cards, equivalent to about 800 petaflops of processing power [4]. The estimated training cost for GPT-4 is around $63 million [3]. These figures indicate that training GPT-4 incurs a significant energy cost.

As for the hardware, GPT-4 is a large multimodal language model with approximately 1.8 trillion parameters across 120 layers, making it 10 times larger than GPT-3. It uses a Mixture of Experts (MoE) model with 16 experts, each having about 111 billion parameters [3]. The model is trained on approximately 13 trillion tokens from various sources, including internet data, books, and research papers [3]. The exact hardware specifications are not publicly available, but the scale of the model suggests that it requires a substantial amount of specialized hardware for training.

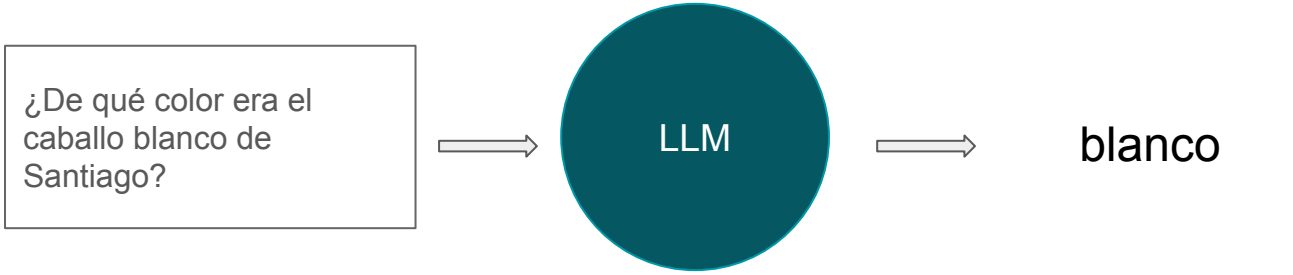
Pongamosle nombre



Prompt

Completion

Pongamosle nombre

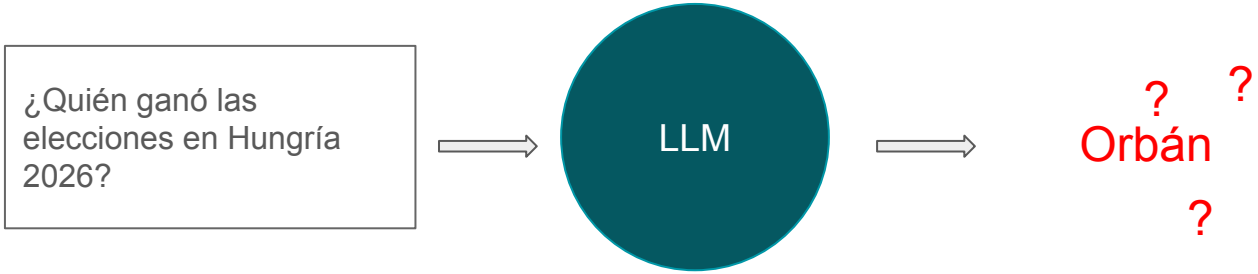


Prompt

Completion

✨Prompt Engineering✨

Pongamosle nombre



Pongamosle nombre

¿Quién ganó las elecciones en Hungría 2026?

LLM

Magyar

Pongamosle nombre

RAG: Retrival Augmented Generation

¿Quién ganó las elecciones en Hungría 2026?

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✨Context Engineering✨

DUDAS